In order to understand the Permutation and Combinations we need a concept called "Factorials" which will be discussed in the next section.

## 4.3 Factorials

Factorial of a natural number $n$ is the product of the first $n$ natural numbers. It is denoted by $n!$. That is,

$$n! = 1 \times 2 \times 3 \times \cdots \times n.$$

We read this symbol as "$n$ factorial" or "factorial of $n$". The notation $n!$ was introduced by the French mathematician Christian Kramp in the year 1808. Note that for a positive integer $n$

$$\begin{aligned} n! &= n \times (n-1) \times (n-2) \times \cdots \times 3 \times 2 \times 1 \\ &= n(n-1)! \quad \text{for} \quad n > 1 \\ &= n(n-1)(n-2)! \quad \text{for} \quad n > 2 \\ &= n(n-1)(n-2)(n-3)! \quad \text{for} \quad n > 3 \quad \text{and so on.} \end{aligned}$$

Observe that,

$$\begin{aligned} 1! &= 1 \\ 2! &= 2 \times 1 = 2 \\ 3! &= 3 \times 2 \times 1 = 6 \\ 4! &= 4 \times 3 \times 2 \times 1 = 24 \\ 5! &= 5 \times 4 \times 3 \times 2 \times 1 = 120 \\ \ldots &= \ldots \\ 22! &= 22 \times 21 \times 20 \times \cdots \times 3 \times 2 \times 1 \quad = 1124000727777607680000 \end{aligned}$$

The number 22 ( the Birth date of Ramanujan) has a special place with respect to factorial that, it is the least integer N greater than 1 whose factorial has exactly N digits.

It will be a good exercise for both students and teachers to find the next number $N$ such that $N!$ has exactly $N$ digits.

Note that $0! = 1$ is evident by substituting $n = 0$ in the equation $(n+1)! = (n+1) \times n!$ as $1! = (0+1) \times 0! \implies 0! = \frac{1!}{1} = 1$. This way, we talk of factorial for non-negative integers. Note that factorials can be extended to certain negative numbers and also to complex numbers, which are beyond the scope of this book.

We shall now discuss certain examples in order to familiarise the computation of factorials.

**Example 4.16** Find the value of

$$\text{(i)}\ 5! \quad \text{(ii)}\ 6! - 5! \quad \text{(iii)}\ \frac{8!}{5! \times 2!}.$$

**Solution:**

(i) $5! = 5 \times 4 \times 3 \times 2 \times 1 = 120.$

(ii) $6! - 5! = 6 \times 5! - 5! = (6-1) \times 5! = 5 \times 120 = 600.$

(iii) $\dfrac{8!}{5! \times 2!} = \dfrac{8 \times 7 \times 6 \times 5!}{5! \times 2!} = \dfrac{8 \times 7 \times 6}{2} = 168.$